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Revolutionizing Stuck Completion Retrieval: A Novel Approach Using Cemented Inverted Tie Back Receptacles

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Abstract

This paper proposes an innovative method to bypass retrieving stuck completions, such as seal assemblies or packers, by utilizing an inverted Tie Back Receptacle (TBR). By doing so, significant rig time and costs associated with traditional fishing and milling operations during well de-completion will be saved.

A structured methodology was developed to deploy the cemented inverted TBR. The operational sequence consisted of the following. First, cut tubing above stuck completion and retrieve it. After that, install a retrievable down hole barrier below the stuck completion and dress the top of the cut joint. Next, perform a dummy run with the inverted TBR to ensure proper seating. Then, run the inverted TBR with either a rotating liner hanger system or casing string to achieve effective isolation of the interval above the stuck completion. Finally, execute a cement job and clean-out runs before completing the well.

Successful deployment of the cemented inverted TBR was achieved in three separate wells, demonstrating its efficacy in overcoming stuck completion challenges. Post-job evaluations revealed excellent isolation of the targeted zones above the stuck completion, confirming the quality of the completed wells. Compared to traditional retrieval methods, this approach showed notable improvements in terms of reduced Non-Productive Time (NPT), enhanced operational efficiency, and optimizing overall costs. Additionally, the cemented inverted TBR proved effective in achieving uniform internal diameters, thereby simplifying future well interventions. Overall, the results indicate that this innovative solution provides a viable alternative to conventional retrieval techniques, offering numerous advantages in terms of both technical feasibility and practical application.

This idea presents a groundbreaking alternative to conventional practices when dealing with stuck completions. By leveraging the concept of inverted TBRs, it introduces a new paradigm shift in addressing these challenges. With successful deployments in multiple wells, this technology offers a proven solution for reducing costs, enhancing efficiency, and improving overall well productivity – ultimately contributing valuable insights to the existing body of knowledge in the oil and gas industry.

Introduction

Workover operations are a critical aspect of the oil and gas industry, aimed at maintaining or restoring the productivity of a well that has experienced a decline in performance. These operations involve a range

of complex and challenging tasks, including well de-completion, which is the process of removing the existing completion to access the wellbore. However, a common challenge faced by the rigs during well de-completion process is stuck or damaged completion equipment, which can require traditional fishing and milling operations. These operations can be time-consuming and costly, resulting in significant rig Non-Productive Time (NPT) and an increased operational cost.

To address this issue, in mechanical workover operations specifically, a novel method has been developed, leveraging the concept of an inverted TBR to bypass stuck completions. This approach has the potential to significantly reduce rig NPT, enhance operational efficiency, and optimize overall costs. The primary objective of this paper is to propose an innovative method for bypassing the retrieval of stuck completions during well de-completion by using an inverted TBR.

Objectives/Scope

The primary scope includes the development of a structured methodology for deploying the inverted TBR and having the option to cement it, as well as the evaluation of its efficacy in overcoming stuck completion challenges. The specific objectives of this study are:

- **Design and Development:** To design and develop a novel approach for bypassing stuck completions retrieval using an inverted TBR, which can be cemented to achieve effective isolation of the interval above the stuck completion. This objective is crucial in addressing the challenges associated with stuck completions, which can lead to significant costs and operational inefficiencies.
- **Step-by-Step Procedure:** To establish a step-by-step procedure for deploying the cemented inverted TBR, including the use of a CCL log, retrievable downhole barriers, pipe cutter, and clean-out runs. To establish a step-by-step procedure for deploying the cemented inverted TBR, including the use of a Casing Collar Locator (CCL) log, retrievable downhole barriers, pipe cutter, and clean-out runs. This procedure will provide a clear and systematic approach to deploying the cemented inverted TBR, minimizing the risk of errors and ensuring a successful operation.
- **Effectiveness Evaluation:** To evaluate the effectiveness of the cemented inverted TBR approach in overcoming stuck completion challenges, including obsolete packers and stuck seal assemblies.
- **Comparison with Traditional Methods:** To compare the performance of the cemented inverted TBR approach with traditional retrieval methods in terms of rig Non-Productive Time (NPT), operational efficiency, and overall costs. This comparison will provide a comprehensive understanding of the benefits and limitations of the cemented inverted TBR approach and its potential to improve operational efficiency and reduce costs.
- **Applicability and Versatility:** To demonstrate the applicability and versatility of the cemented inverted TBR approach in different well scenarios, including wells with varying completion configurations and geological conditions. This demonstration will provide evidence of the cemented inverted TBR approach's ability to adapt to different well environments and its potential to be used in a wide range of applications.

The scope of this manuscript also includes the analysis of the results from the successful deployments of the cemented inverted TBR approach in three separate wells, with a focus on the lessons learned, best practices, and areas for future improvement. Additionally, this study aims to provide a comprehensive review of the existing literature on stuck completion challenges and retrieval methods, highlighting the limitations and drawbacks of traditional approaches and the potential benefits of the proposed innovative method.

Workover Operations

Oil and gas workover are critical process in the oil and gas industry that involves the repair, maintenance, or modification of a well to improve its production or injection performance. It can be categorized into two main types: re-entry or mechanical workover.

Re-Entry Workover involves sidetracking the previously drilled and abandoned lateral to restore or improve its production. The goal of a re-entry workover is to revive a non-producing or underperforming well, often by drilling new sections and applying new technologies or techniques that were not available when the well was initially drilled. Re-entry workovers can be used to access new reservoirs, bypass damaged or plugged zones, or to improve the well's overall production performance.

Mechanical workover, on the other hand, focuses on repairing, maintaining equipment, or addressing mechanical issues that affect the well's productivity. During mechanical workover planning phase, the well completion will be evaluated to assess if the well productivity can be restored. There are two types of completion. Completions connect the reservoir to the well, as shown in [figure 1](#), or upper completion that act as conduit from the reservoir to the surface facilities, as shown in [figure 2](#). The majority of mechanical workover issues such as high TCA, tubing leak, or tubing obstruction can be addressed by replacing the existing upper completion. As a result, the retrieval of upper completion will be required to be replaced.

The most common upper completion includes retrievable production packers to isolate the production zone from the rest of the wellbore while providing a seal between the tubing and casing. In addition, it will include a seal assembly to connect the tubing to Polished Bore Receptacle (PBR) above the packer. This system allows the retrieval of seal assembly without packer retrieving. Despite designing these packers and seal assembly to be retrieved, in many cases the seal assembly will be stuck leading to costly milling and fishing operations.

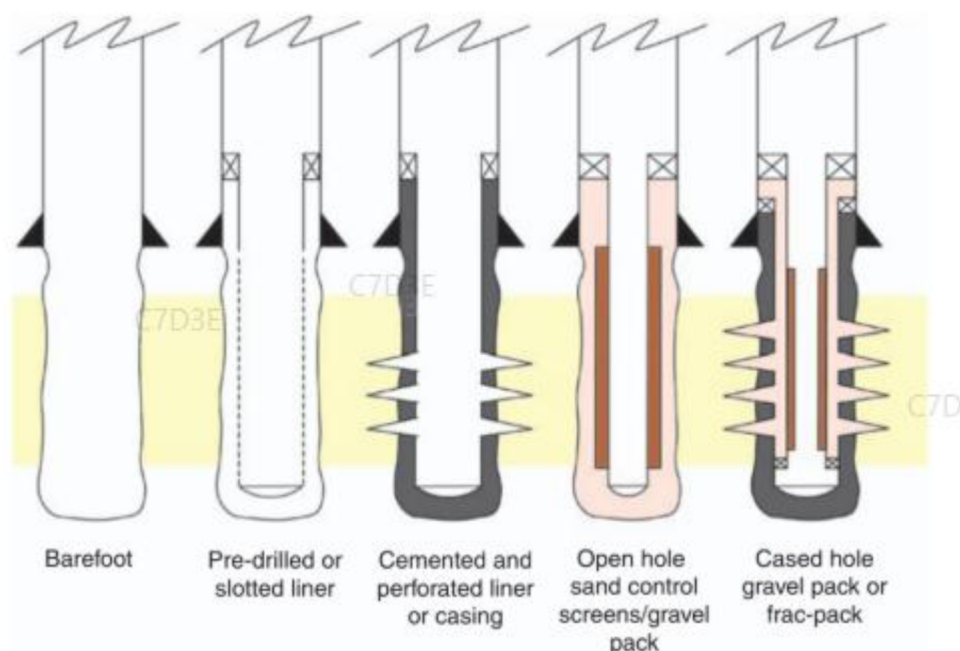


Figure 1—Reservoir Completion Methods

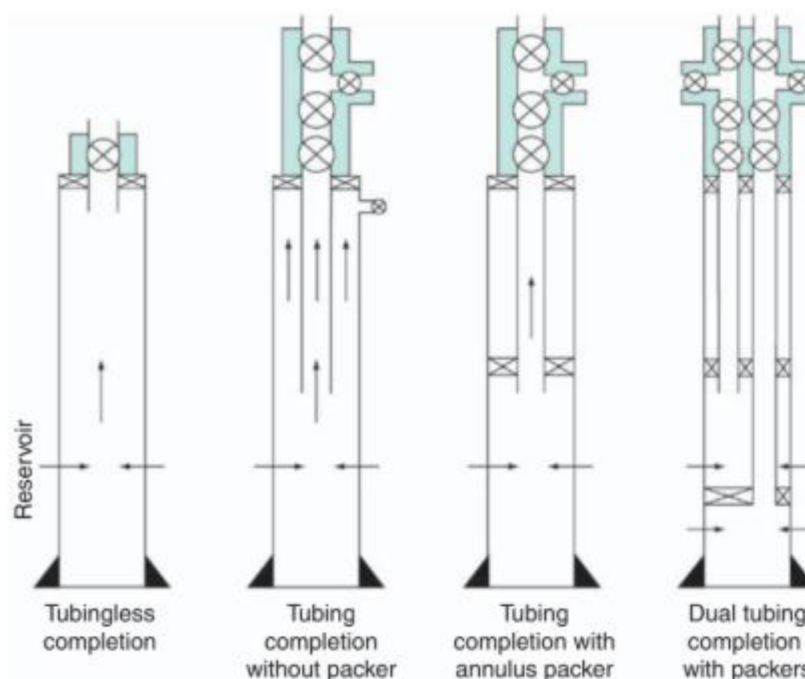


Figure 2—Upper completion methods

Methods, Procedures, and Process

To avoid the fishing and milling operation, A structured methodology was developed to deploy the cemented inverted TBR. The operational sequence consisted of the following steps:

1. **CCL Log Deployment:** A Casing Collar Locator (CCL) log must be deployed to locate the middle of the full joint above the stuck completion to ensure avoiding the casing collars. This will allow the inverted TBR to swallow and overlap the cut joint from the old completion smoothly.
2. **Retrievable Downhole Barrier:** A retrievable downhole barrier below the stuck completion will be run. In addition to the drilling mud used in the hole, this will ensure having full control of the well and avoid any risk of having formation kicks. Overall, the drilling mud is considered as the first object preventing formation kicks or as known "primary barrier" as shown in [figure 2a](#). However, during the inverted TBR deployment and cement operations, the well might encounter losses which will introduce new challenges. If a retrievable downhole plug is not available, a TTBP or drillable bridge plug can be deployed to maintain the reservoir isolation. Drilling the drillable bridge plug is a smoother operation compared to drilling cement or having a well control situation.
3. **Sand Dumping:** A sand will be dumped above the retrievable downhole barrier to avoid the cement or any junk accumulation above the plug as shown in [figure 3](#). Comparing the sand to the cement, the sand can be easily cleared after the operation by circulating the hole clean with high rate just above the sand. On the other hand, the cement will require running with a bit and attempting to drill the cement and for worst drill the plug since the retrieving profile is cemented and the retrieving tool can't be used.
4. **Pipe Cutting:** A pipe cutter will be run to cut inside the old completion tubing and allow retrieving the free tubing, [figure 4](#). There are many types that can be used based on inclination and well geometry. The most common tools are Guillotine Cutter, Hydraulic Pipe Cutter, Chemical Cutter, and Explosive Cutter. Guillotine Cutter is a mechanical pipe cutter that uses a blade to cut the pipe. Hydraulic pipe cutter uses hydraulic pressure to drive the blade and shear the pipe. This type is usually used for cutting larger diameter pipes. Chemical cutter uses a chemical reaction to cut the pipe. Usually it's used in

high pressure and temperature environment. Lastly, explosive cutter uses explosive charges to cut the pipe. this method used for large diameter pipe where the other cutting methods are not feasible.

5. **Clean-Out Run:** A clean-out run with a mill will be performed, if required, to dress the top of the cut joint, [figure 5](#). This clean-out run is based on the tubing and well condition after retrieving the cut tubing as well as the condition of the cut. If obstruction is expected in the overlapping interval above the stuck completion and old casing annulus, a special mill such as washover mill will be run to allow accessing the desired overlapping area since it has a large diameter thin walled pipe to clear it from any obstructions.
6. **Dummy Run:** A dummy run with the inverted TBR will be run to ensure proper seating above the old cut tubing, [figure 6](#), with the inverted TBR as shown in [figure 7](#). Prior engaging to the old completion, the rig can normal circulate through the BHA internal and through the inverted TBR while monitoring the pump pressure and carefully attempting to engage. If required, a slight rotation can be applied to smoothen the engagement. Once the pressure increases, the TBR should be engaged to the old completion. At this point, all the parameters should be recorded to identify the trend and compare it with the proposed next run.
7. **TBR Engagement:** Once a proper seating of the inverted TBR is ensured, the TBR can be retrieved to be run with a rotating liner hanger system or casing to achieve effective isolation of the interval above the stuck completion, ([figure 8](#)). The liner hanger and casing systems can be used if cement operation is needed. On the other hand, another option to run the inverted TBR with tubing and packer completion if the area above the stuck completion is not required to be isolated with cement.
8. **Cement Job:** The cement job should be executed to ensure proper isolation of the interval above the stuck completion, ([figure 9 and 10](#)). The cement will act as mechanical barrier to protect the casing above the stuck completion and prevent any crossflow in the future.
9. **Clean-Out Runs:** Clean-out runs will be performed with a CP mill before completing the well to clean any excess cement inside the rotating liner hanger or casing and ensure access inside the old completion. In this clean out trip, after cleaning all the cement, the BHA will be lowered above the sand located above retrievable plug and the rig will circulate with high rate to circulate all the sand out of the hole. This will enable running with downhole barrier retrievable tool and enable retrieving the plug.
10. **Downhole Barrier Retrievable:** The bridge plug retrieving tool will be run in the well to latch into the retrievable bridge plug profile and the packer will be retracted to allow plug retrieving.
11. **Complete the well,** if needed, an upper completion such as tubing with annulus packer can be deployed ([figure 11](#)). This upper completion will isolate the production zone from the rest of the wellbore while providing a seal between the tubing and casing.

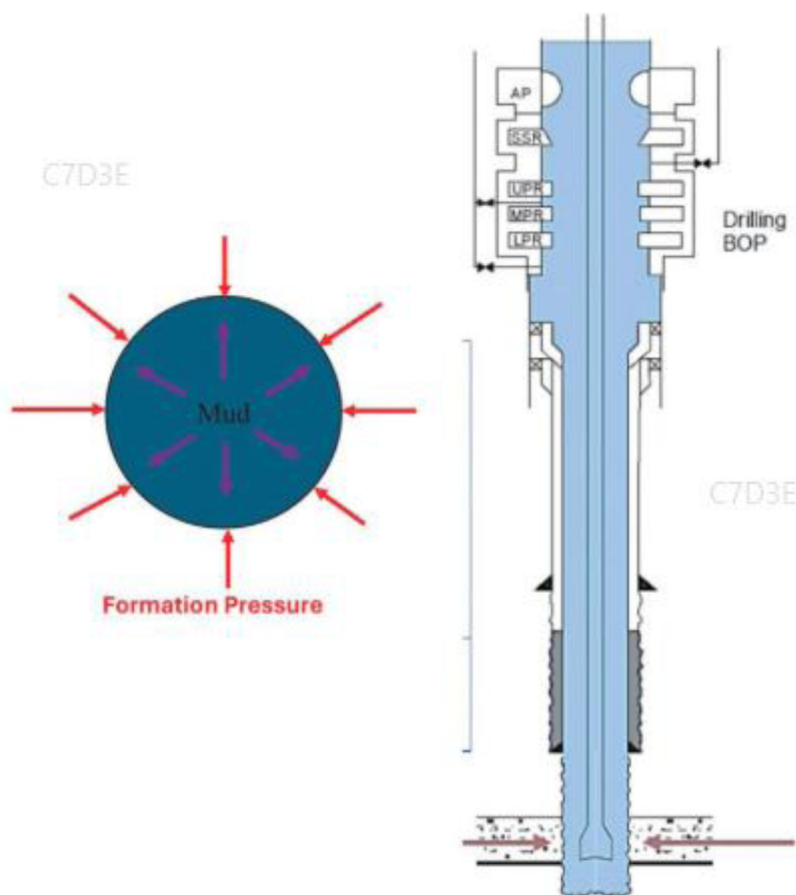


Figure 2a—Drilling mud force direction in a wellbore

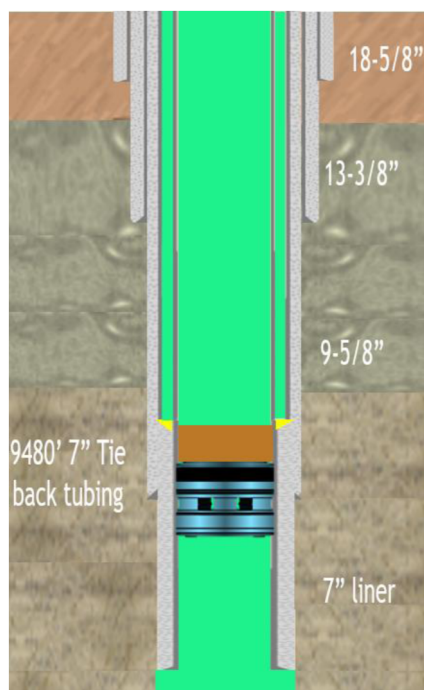


Figure 3—Set RBP below stuck seal assembly to isolate reservoir and capped it with sand.

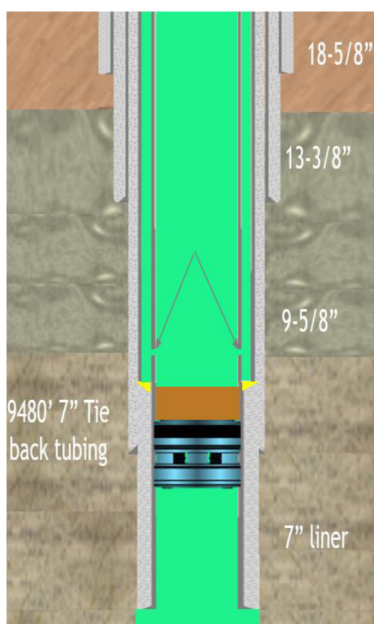


Figure 4—Tubing was cut above stuck seal assembly.

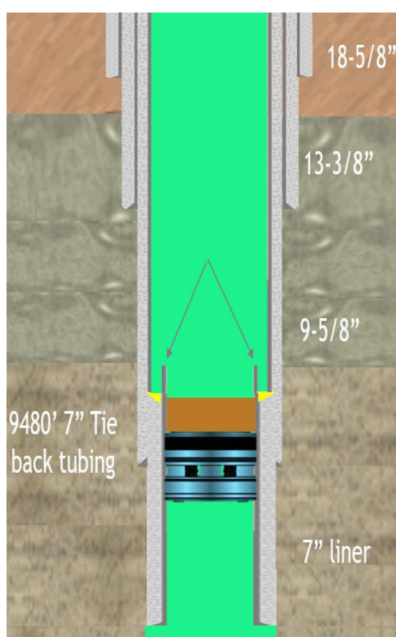


Figure 5—Top of cut-off joint was dressed and ready for TBR deployment.



Figure 6—Top of Cut-off Joint.

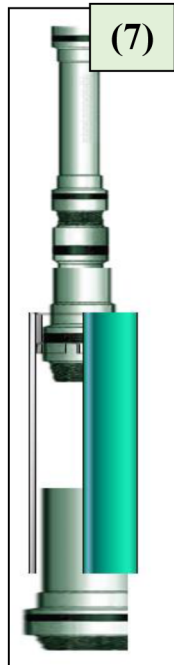


Figure 7—Performed Dummy run with Inverted TBR.

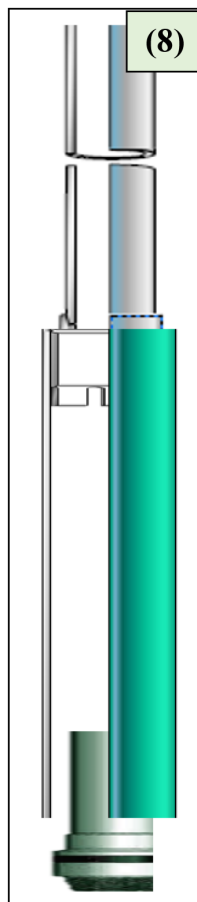


Figure 8—Swallowed Top of Cut-off Joint with Inverter TBR, casing and Liner Hanger

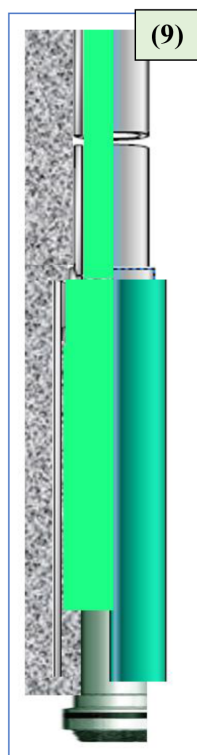


Figure 9—Performed cementing job.

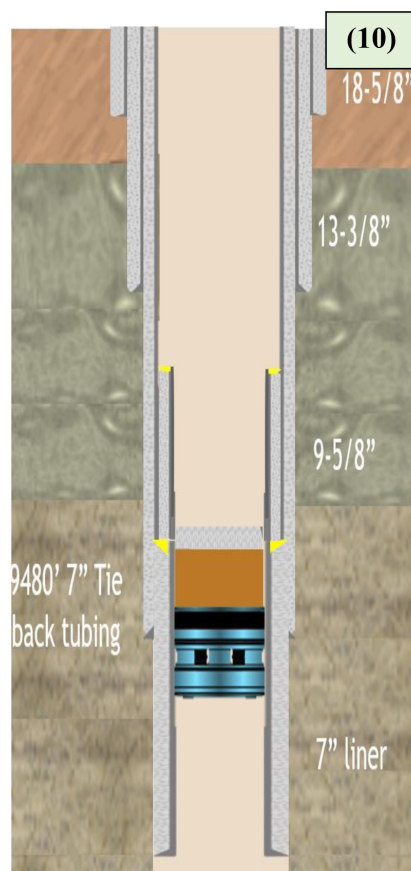


Figure 10—Tie back cemented liner with inverted TBR.

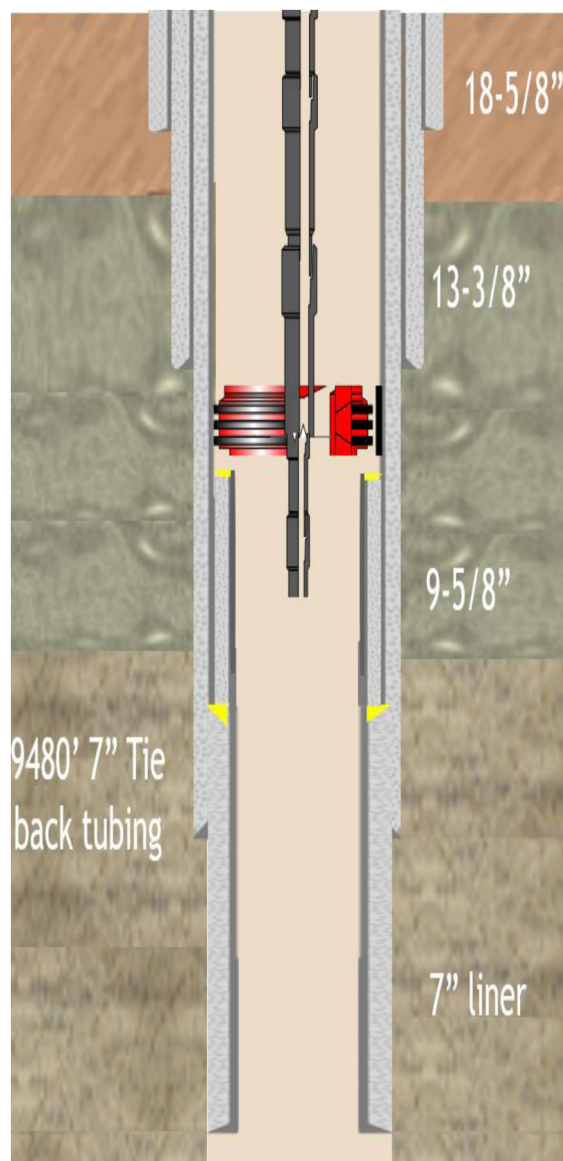


Figure 11—Completed the well with tubing and packer.

Results

This approach was implemented with different challenges. Most common challenges are due to pre-installed obsolete packers that couldn't be retrieved, and encountering a stuck seal assembly in the top of the liner hanger polish bore receptacle. To maintain the well quality and well integrity by not compromising accessibility to the reservoir, the cemented inverted TBR approach was applied without problems. These deployments were considered successful deployments of the cemented inverted TBR, demonstrating its efficacy in overcoming stuck completion challenges.

Post-job evaluations revealed excellent isolation of the targeted zones above the stuck completion, confirming the quality of the completed wells. Compared to traditional retrieval methods, this approach showed notable improvements in terms of reduced rig NPT, enhanced operational efficiency, and optimized overall costs by avoiding fishing and milling operations. Additionally, the cemented inverted TBR proved effective in achieving uniform internal diameters, thereby simplifying future well interventions.

Discussion

The cemented inverted TBR approach has been successfully deployed in three wells, demonstrating its efficacy in overcoming stuck completion challenges. The approach has shown notable improvements in terms of reduced rig NPT, enhanced operational efficiency, and optimized overall costs. The use of a cemented TBR provides a reliable and durable barrier against fluid migration and pressure transmission, which can help to prevent well control issues and minimize the risk of environmental damage.

The cemented inverted TBR approach has several advantages over traditional retrieval methods. Firstly, it eliminates the need for fishing and milling operations, which can be time-consuming and costly. Secondly, it provides a reliable and durable barrier against fluid migration and pressure transmission, which can help to prevent well control issues and minimize the risk of environmental damage. Thirdly, it achieves uniform internal diameters, thereby simplifying future well interventions.

However, the cemented inverted TBR approach also has some limitations. Firstly, it requires a high degree of precision and accuracy in the deployment of the TBR, which can be challenging in certain well environments. Secondly, it may not be suitable for all types of stuck completions, and the suitability of the approach will depend on the specific well conditions and completion configuration.

Conclusion

In conclusion, the cemented inverted Tie Back Receptacle (TBR) approach has been successfully demonstrated as a viable alternative to traditional retrieval methods for overcoming stuck completion challenges in oil and gas wells. The results of this method show that this innovative approach can significantly reduce rig Non-Productive Time (NPT), enhance operational efficiency, and optimize overall costs. The use of a cemented TBR provides a reliable and durable barrier against fluid migration and pressure transmission, which can help to prevent well control issues and minimize the risk of environmental damage. Additionally, the approach achieves uniform internal diameters, simplifying future well interventions.

The successful deployment of the cemented inverted TBR demonstrates the efficacy of this approach in overcoming stuck completion challenges. Compared to traditional retrieval methods, this approach showed notable improvements in terms of reduced rig NPT, enhanced operational efficiency, and optimized overall costs. The elimination of fishing and milling operations, which can be time-consuming and costly, is a significant advantage of the cemented inverted TBR approach.

The cemented inverted TBR approach also offers several other benefits, including improved well integrity and reduced risk of well control issues. The use of a cemented TBR provides a reliable and durable barrier against fluid migration and pressure transmission, which can help to prevent well control issues and minimize the risk of environmental damage. Additionally, the approach achieves uniform internal diameters, simplifying future well interventions and reducing the risk of future completion challenges.

While the approach may have some limitations, such as requiring high precision and accuracy in deployment, it has been shown to be effective in a variety of well scenarios. The suitability of the approach will depend on the specific well conditions and completion configuration, and further work is needed to refine and optimize the approach. However, the results of this approach demonstrate the potential of the cemented inverted TBR approach to revolutionize the way stuck completion challenges are addressed in the oil and gas industry.

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